

Program 5:

Demonstrate CI Integration by Running Jenkins in a Docker Container

Project Structure:

program_5/

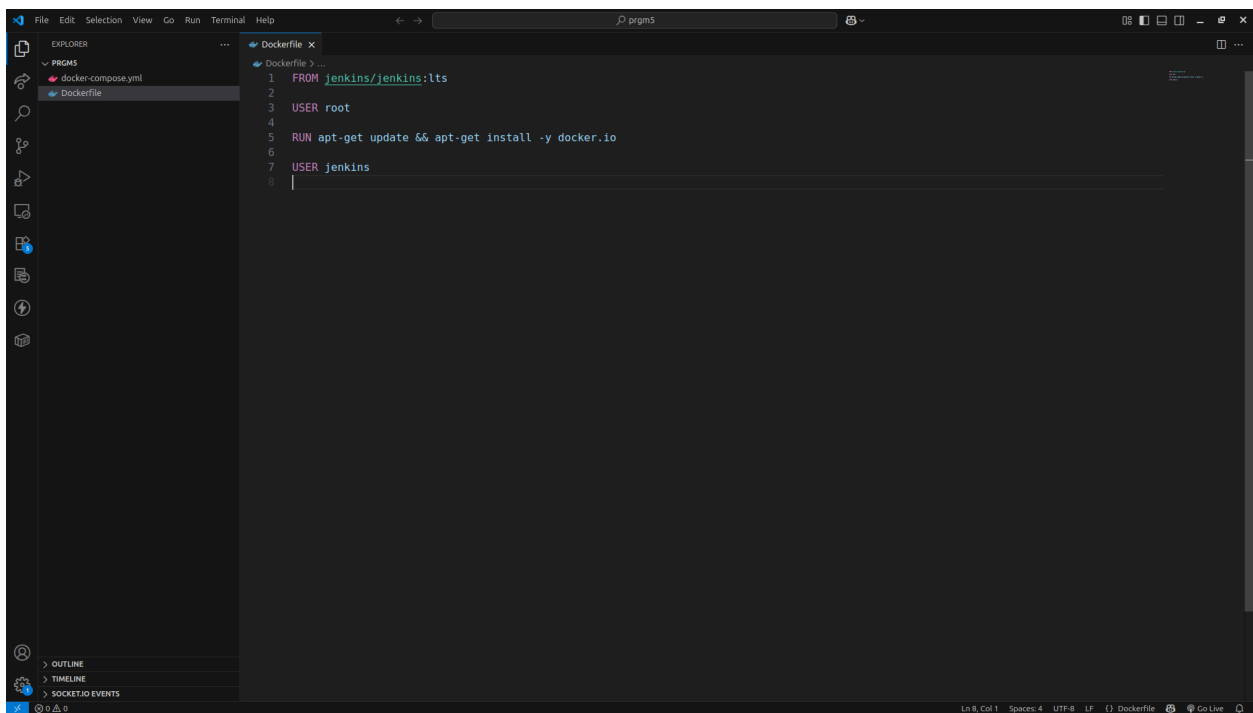
—>Dockerfile

—>docker-compose.yml

Procedure:

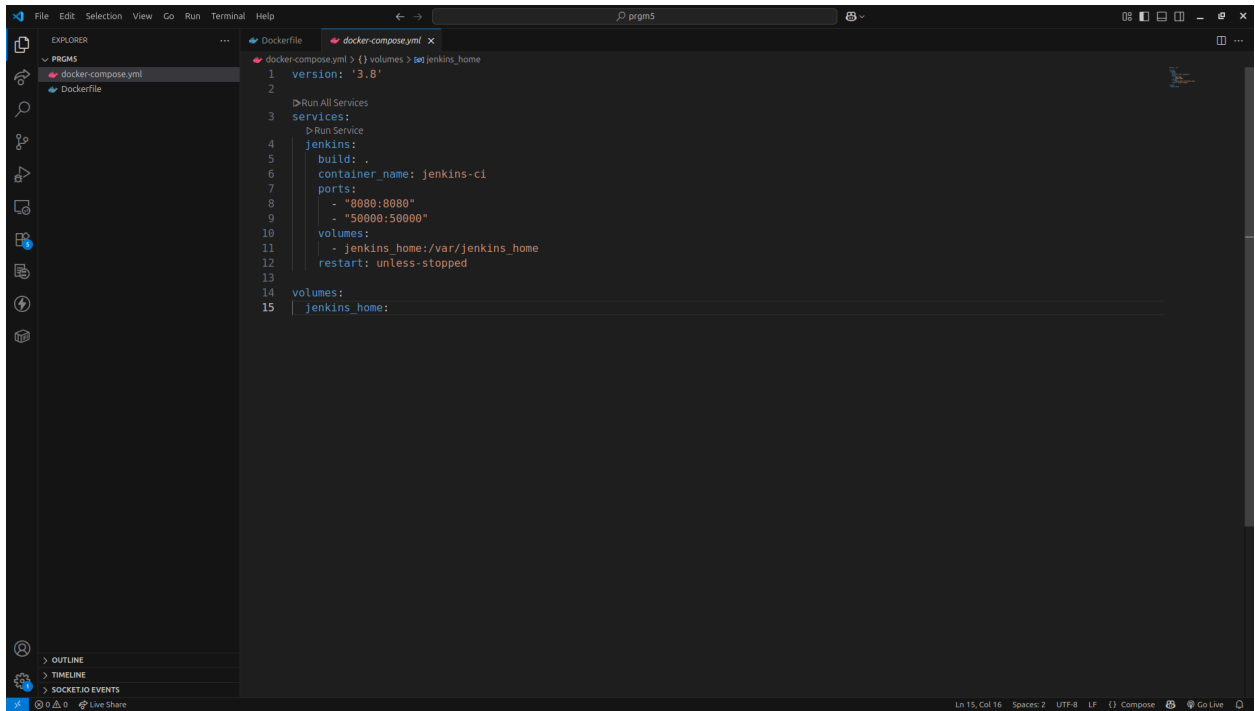
Create a folder (ex: program_5)

Step 1: Create a Dockerfile without any extensions and add the following content



```
1 FROM jenkins/jenkins:lts
2
3 USER root
4
5 RUN apt-get update && apt-get install -y docker.io
6
7 USER jenkins
8
```

Step 2: Create a docker-compose.yml file with following code
nano docker-compose.yml



```
1 version: '3.8'
2
3 services:
4   jenkins:
5     build: .
6     container_name: jenkins-ci
7     ports:
8       - "8080:8080"
9       - "50000:50000"
10    volumes:
11      - jenkins_home:/var/jenkins_home
12    restart: unless-stopped
13
14 volumes:
15   jenkins_home:
```

Step 3: Build an image
docker build -t program_5 .
Run the container
docker run -p 8080:8080 program_5

Step 4: Verify localhost details on browser
<http://localhost:8080>